

SecureSync

-Ag19

About Us

SecureSync focuses on solving real-world security problems faced by manufacturing companies with multiple plants. Our system ensures that only trusted plants can send data, every payload is verified, and all security events are visible through a centralized dashboard.



Our Solution



Zero-Trust Model

Every request is treated as untrusted.

Verified Plants Only

Only authenticated sources can send data.

End-to-End Security

Comprehensive protection throughout data transmission.

SCALABILITY: SecureSync scales horizontally while preserving security, visibility, and performance in real time

Admin Panel

Manage security configuration, review audit logs, and monitor transmissions.

Open Admin

User Panel

Send secure data to the server with encryption and digital signatures.

Open User

What Do We Have?

Admin Panel

Manage security settings, monitor audit logs, control IP access, and handle blocked or rejected requests in real time.

User Panel

Securely send encrypted and digitally signed data to the server with instant verification and response.

If you want it even shorter (1 line each) or more technical, say the word

User Panel (Main)

This screen allows an authorized plant to securely transmit data to the server.

The user selects a plant ID, sets data priority, and submits the payload.

Before transmission, the data is digitally signed and encrypted.

The server then performs IP whitelist, rate-limit, signature verification, and decryption checks.

Successful validation is confirmed with real-time logs and visual status feedback.

 **Send Secure Data**  

Select Plant ID

Plant-A (127.0.0.1) 

Data Priority

Non-Critical (Standard) 

Data Payload

Data Is Typed Here... 

 Sign, Encrypt & Send

 Data successfully verified and accepted by server!

[02:50:47] Initiating transmission from Plant-A (127.0.0.1)...

[02:50:47] Signing data with Plant-A Private Key...

[02:50:47] Encrypting payload (AES-256-GCM)...

[02:50:47] Transmission package ready. Sending to Server...

[02:50:47] Server received request. Performing Security Checks...

[02:50:47]  IP Whitelist & Rate Limit Checks PASSED.

[02:50:47] Verifying Digital Signature & Decrypting...

[02:50:47]  Signature VERIFIED. Data Integrity Confirmed.

[02:50:47] Decryption Successful. Data stored.

Technical & Clean

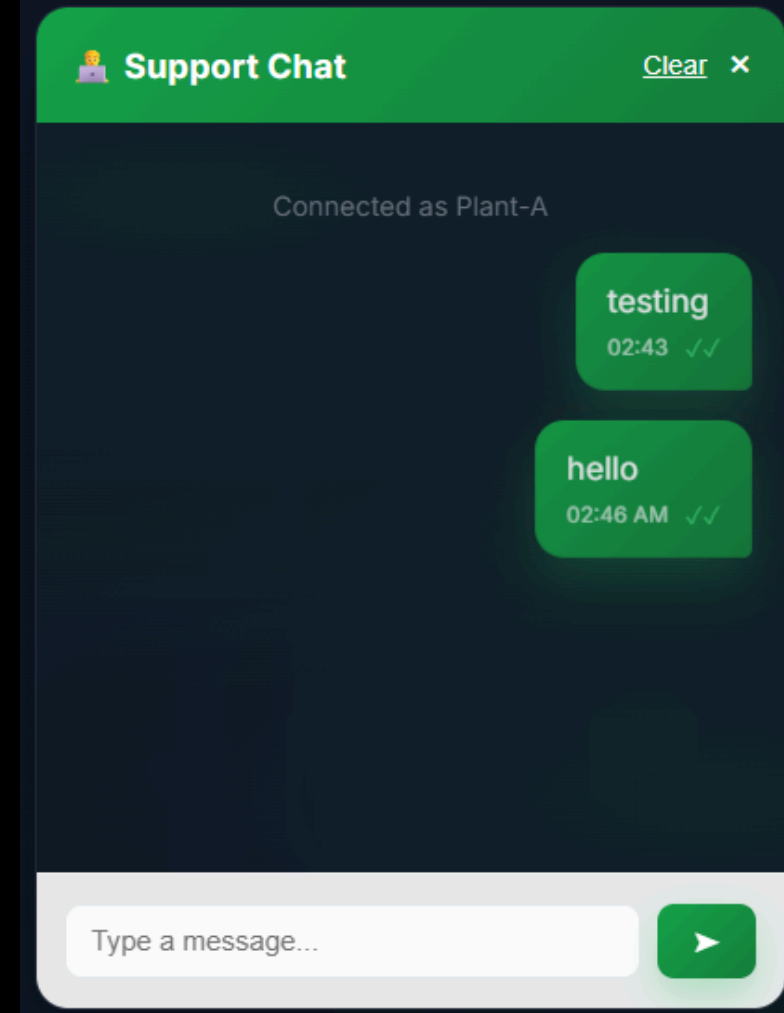
Security Protocols & Rules

- ✓ **Whitelist Policy:** Only authorized IP addresses can transmit data. Unauthorized IPs are instantly rejected.
- ⚠ **Non-Critical Rate Limit:** Maximum **5 requests** per minute. Exceeding this will result in a temporary block.
- 🔔 **Critical Rate Limit:** Maximum **10 requests** per minute. High-priority data has higher throughput allowances.
- 🚫 **Blocking Policy:** Repeated violations (3+ rejections) may lead to a permanent IP block requiring admin intervention.

SecureSync v1.0 • Protected by AES-256 & RSA-2048

1 Defines the core security policies of SecureSync, including IP whitelisting, request rate limits for standard and critical data, and automatic blocking of repeated violations—ensuring only trusted and controlled communication.

2 Provides a secure real-time communication channel between plants and administrators. Enables instant messaging, message tracking, and quick issue resolution during data transmission or security events.



Admin Panel



The Admin Panel provides centralized control over SecureSync's security and monitoring system.

Administrators can review real-time audit logs, manage IP whitelists, track accepted and rejected requests, and respond to potential security threats through live monitoring and access controls.

Admin Access

Please authenticate to continue

Password

Login

[← Back to User Panel](#)

TOTAL REQUESTS
203

ACCEPTED
161

REJECTED
39

BLOCKED
2

Refresh Logs

Blocked IPs

Messages

Search

Clear

Audit Log Viewer

Plant ID	IP Address	Time	Status	Reason
Plant-B	192.168.1.10	02:51	REJECTED	IP not whitelisted
Plant-A	127.0.0.1	02:50	ACCEPTED	Valid Signature & Whitelisted
Plant-A	127.0.0.1	02:09	ACCEPTED	Valid Signature & Whitelisted
Plant-A	127.0.0.1	01:43	ACCEPTED	Valid Signature & Whitelisted
Plant-B	192.168.1.10	01:43	REJECTED	IP not whitelisted
Plant-A	127.0.0.1	01:43	ACCEPTED	Valid Signature & Whitelisted
Plant-A	127.0.0.1	01:16	ACCEPTED	Valid Signature & Whitelisted

Attack Monitor

Plant ID	IP Address	Reason
Plant-B	192.168.1.10	IP not whitelisted
Plant-B	192.168.1.10	

Admin Features



 Profile

 User Panel (Send Data)

 Plant Config

 Change Password

 Add Whitelist IP

 Remove Whitelist IP

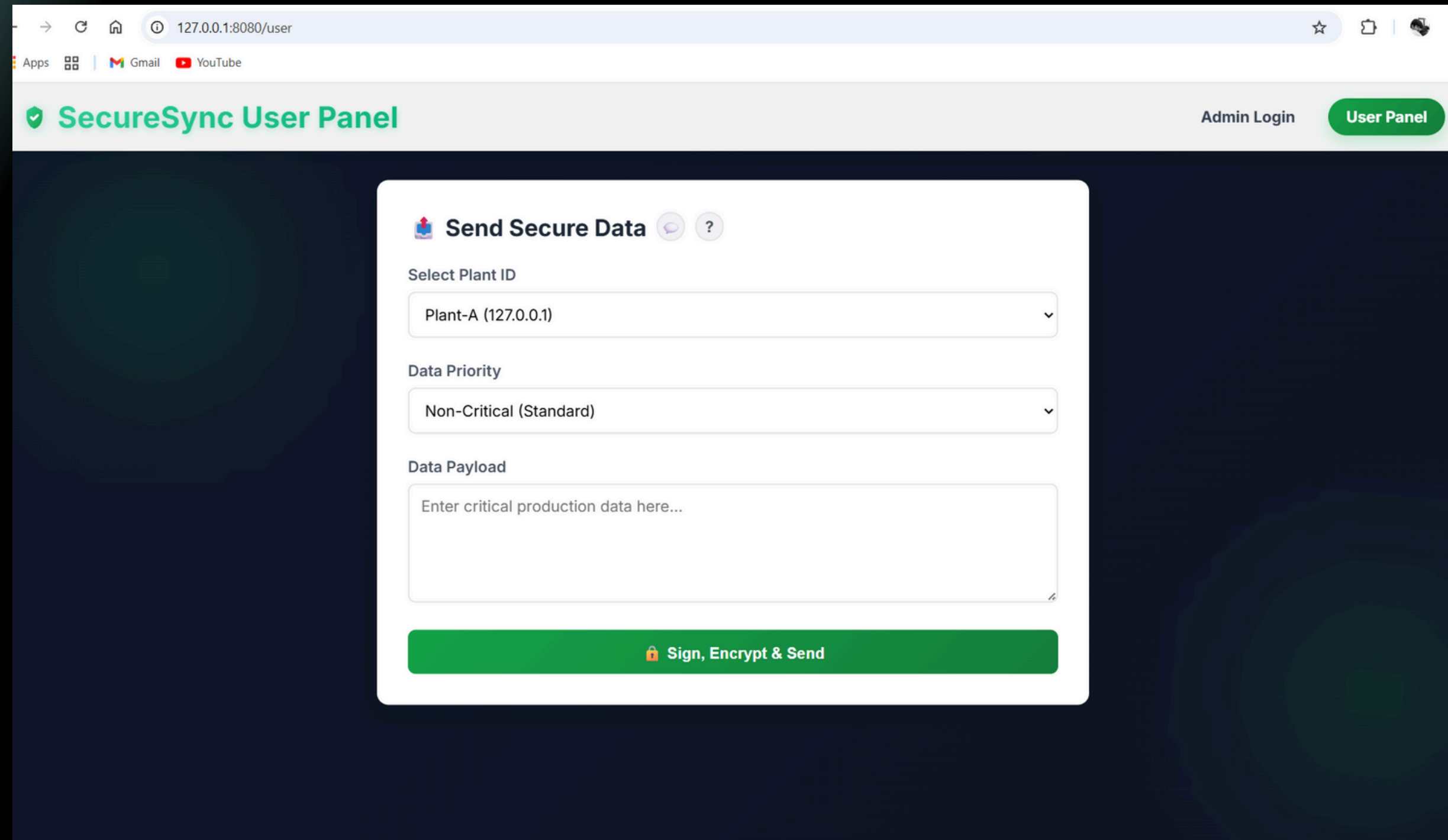
 Sign Out

Admin Features



User Panel (Send Data)

Securely send encrypted and digitally signed data to the server with real-time verification and status feedback.



The screenshot shows a web browser window with the address bar displaying "127.0.0.1:8080/user". The browser's address bar includes navigation icons (back, forward, home, refresh) and a search icon. Below the address bar, there are links for "Apps", "Gmail", and "YouTube". The main content area has a header bar with the "SecureSync User Panel" title and a green checkmark icon. On the right side of the header, there are links for "Admin Login" and "User Panel". The "User Panel" link is highlighted with a green background. The main content area features a central white card titled "Send Secure Data" with a red location pin icon and a question mark icon. The card contains three sections: "Select Plant ID" with a dropdown menu showing "Plant-A (127.0.0.1)", "Data Priority" with a dropdown menu showing "Non-Critical (Standard)", and "Data Payload" with a text input field containing the placeholder text "Enter critical production data here...". At the bottom of the card is a large green button with a lock icon and the text "Sign, Encrypt & Send".

Admin Features



⚙️ Plant Configuration

Configure plant-level security settings such as encryption, digital signature enforcement, and whitelist status.

⚙️ Plant Security Config

Select Plant

Plant-A (127.0.0.1) ▾

IP Whitelisted: True

☒ Assign Encryption

☒ Assign Digital Signature

Save Configuration

Back to Dashboard

Admin Features



Change Password

Update account credentials to maintain secure access and prevent unauthorized usage.

Change Password

Old Password

New Password

Confirm New Password

Update Password

Back to Dashboard

Admin Features

— Remove Whitelist IP

Revoke access for a plant or IP address to immediately block further communication. OR

+ Add Whitelist IP

Authorize a new plant or IP address to allow secure data transmission to the system.



— Remove Whitelist IP

Enter IP Address or Plant Name to remove

Remove

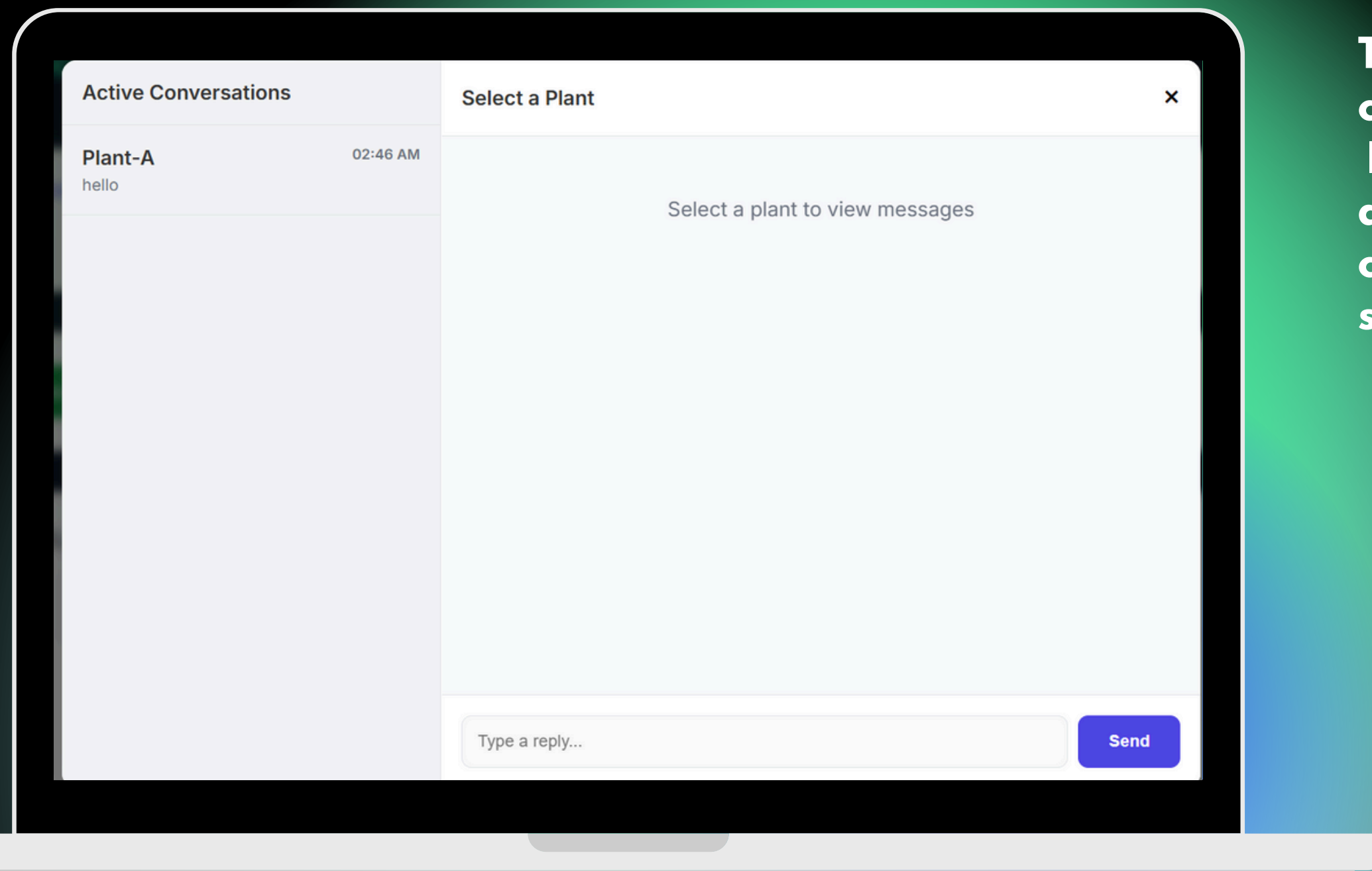
← Back to Dashboard

+ Add Whitelist IP

Enter IP Address or Plant Name (e.g., Plant-A)

Add

← Back to Dashboard



The User Support module enables real-time communication between plants and administrators. It helps resolve data transmission issues, security alerts, and configuration problems instantly through a secure chat interface, ensuring smooth and uninterrupted system operation.

User Support

Without SecureSync



Unverified Data Risk

Data is often trusted blindly, risking security.

Attacks Go Unnoticed

Lack of monitoring allows attacks to proliferate.

No Real-Time Visibility

Malicious actions can remain hidden indefinitely.

Without SecureSync



Data Tampering

**Compromised data
can lead to severe
consequences.**

Production Failures

**Disruptions can halt
operations and cause
losses.**

Security Breaches

**Breaches result in
financial and
reputational damage.**

Tech Stack Overview



Frontend: Web-based dashboard for inputs, visualization, and admin control

Backend: Secure server handling requests, validation, and computation

Security Layer: Encryption for data-in-transit Authentication & access control
Rate limiting and request validation

Logic Engine: Computes safest distance using threat-aware rules

Monitoring & Logs: Real-time logs, audit trails, and admin visibility

Thank You

